CNENGR202 Instrumentation and Control

Part 2 – L2 Pressure sensing

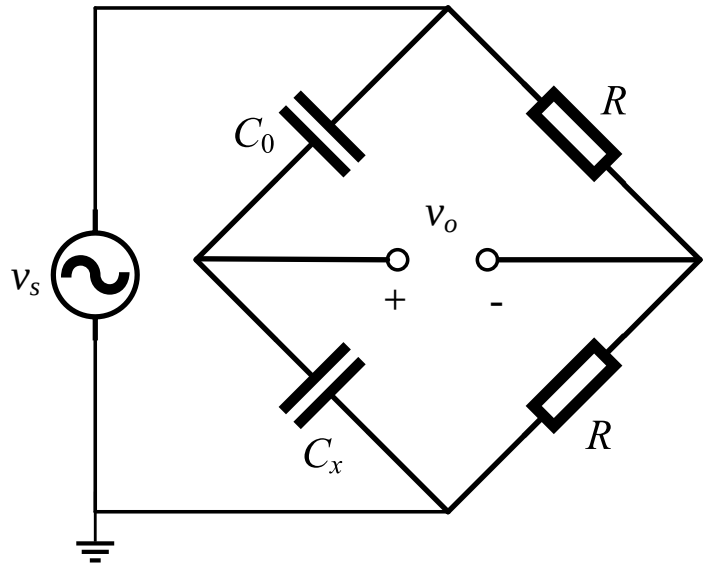
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Class question 1

Use a capacitive humidity sensor within a bridge circuit. How does your voltage output vary with capacitance and therefore humidity?



Solution

$$v_o = \left(\frac{1/sC_0}{1/sC_x + 1/sC_0} - \frac{R}{R+R} \right) v_s = \left(\frac{1}{1+\frac{C_0}{C_x}} - \frac{1}{2} \right) v_s$$

Solution continued

$$C_0 = \frac{\varepsilon_0 \varepsilon_r S}{d} \qquad C_x = \frac{\varepsilon_0 \varepsilon_{rx} S}{d}$$

$$v_o = \left(\frac{1}{1 + \frac{\varepsilon_r}{\varepsilon_{rv}}} - \frac{1}{2} \right) v_s \longrightarrow \text{Humidity}$$

Pressure (P)

- The force per unit area applied to an object in a direction perpendicular to the surface.

1 Atm

101,325 Pascals (Pa) (Nm^{-2})

- Gauge pressure relative to ambient air pressure (e.g tire, blood pressures)
- Absolute pressure relative to vacuum (e.g atmospheric, altimeters, depth)
- Differential pressure is difference in pressure between two points

Absolute pressure in water

- In water, the absolute pressure (where h is depth in metres, and P_0 is the pressure at the surface) is

When h=10m

$$P_h \approx P_0\left(1+h/10\right) \qquad P_h \approx 2P_0$$

- 1 meter down – 1.1 atm
- The Titanic 3750m – 376 atm
- Average Atlantic Ocean depth 3926m – 394 atm
- Deepest part of Pacific ocean 10,916m – 1093 atm



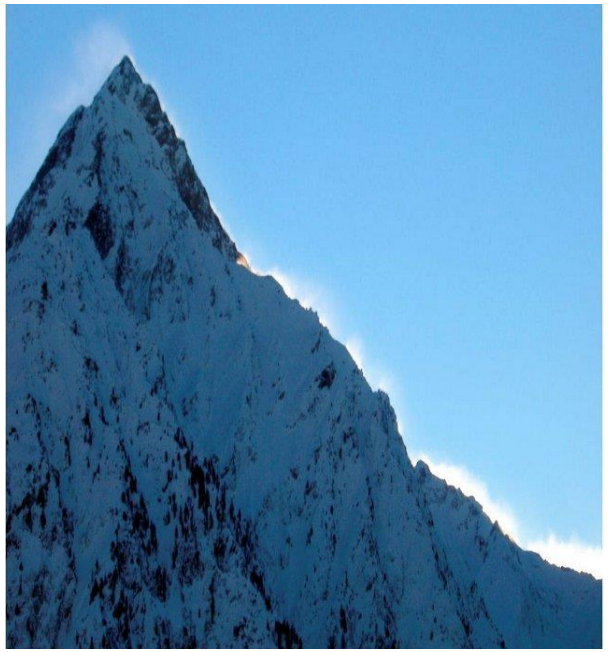
Pressure decreases as altitude increases as

$$P_h = P_0 e^{-mgh/KT} \approx P_0 e^{-0.0001138*h}$$

When h=6100m

$$P_h \approx 0.5 P_0$$

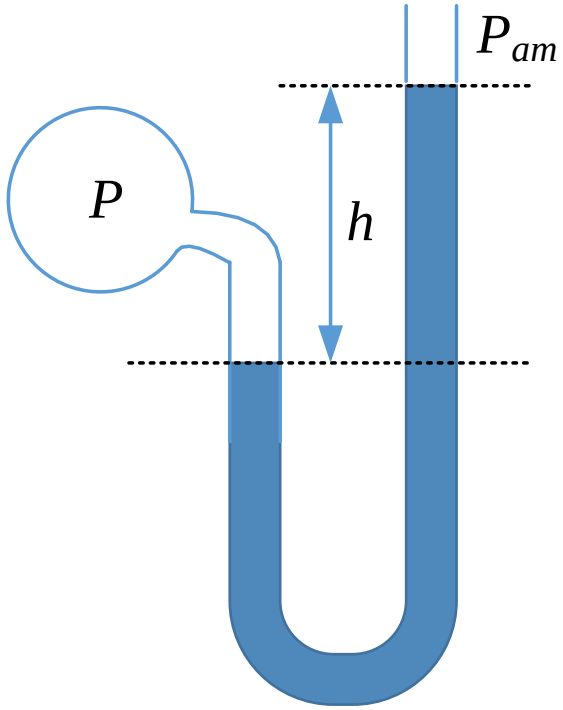
- m is the mass of one molecule in kg (air ~ 4.7×10^{-26} kg)
- g is gravity (9.81 $m.s^{-2}$)
- h is the height (m)
- K is boltzmanns constant (1.38×10^{-23} $J.K^{-1}$)
- T is Temperature (~300K)



Manometer

- Measures pressure by noting ability to displace a column of liquid.
- Pressures often expressed as a depth of a particular fluid, often millimeters of water (ρ = 1,000 kg/m^3) or mercury (ρ = 13,534 kg/m^3).

$$P - P_{am} = \rho g h$$



ρ = liquid density (kg m^{-3})

P = Pressure inside balloon (Pa)

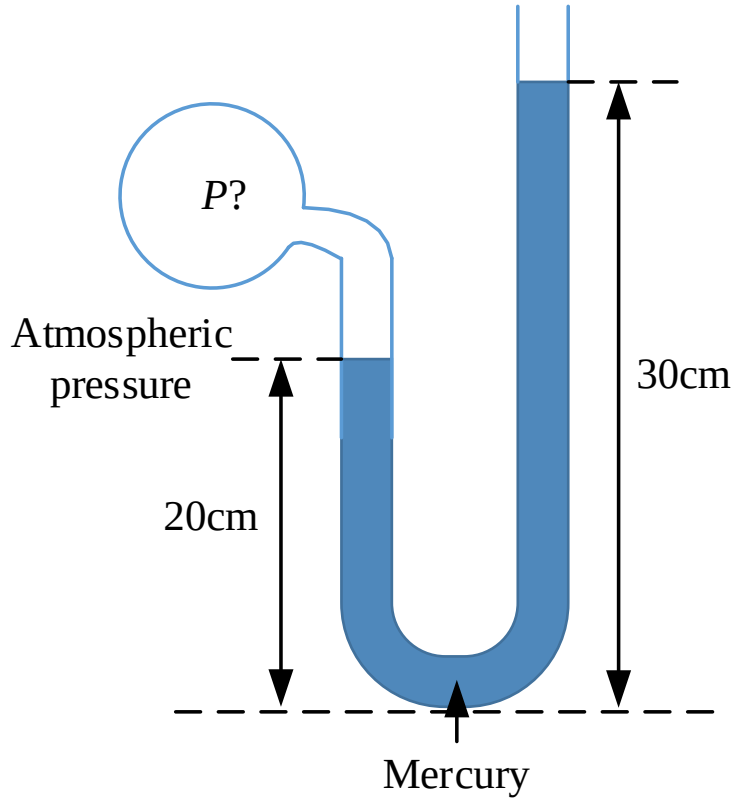
P_{am} = Ambient Pressure (Pa)

g = gravity (m s^{-2})

h = difference between water levels (m)

In class question 1

- What is the pressure *P* of mercury inside the balloon?



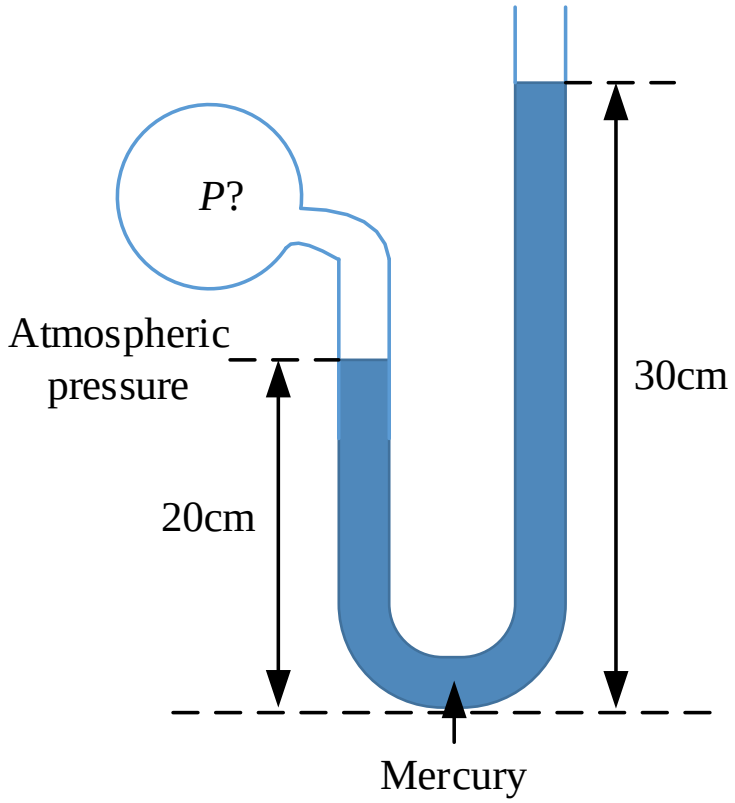
Solution for class question 1

- What is pressure P of mercury inside the balloon?

$$P - P_{am} = \rho g h$$

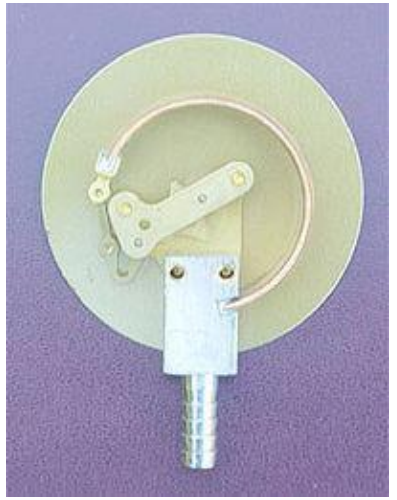
$$P - 101,325 = 13.534 * 9.81 * (0.3 - 0.2)$$

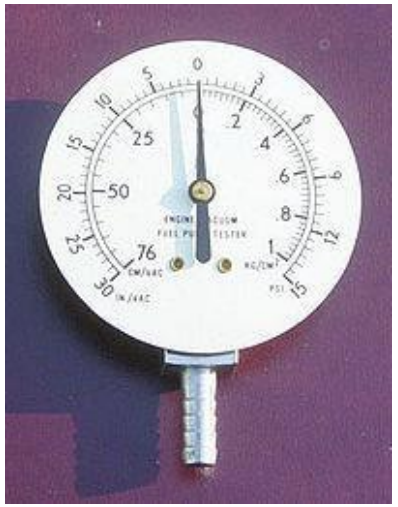
$$P = 13,277 + 101,325 = 114,602 \text{ Pa}$$



Bourdon gauge

- Contains a C-shaped tube, closed at one end.
- When pressure inside the tube increases, it uncurls slightly, causing a small movement at its closed end.
- Levers and gears magnify this and turn a pointer. Fitted to cylinders of compressed gas used in industry and hospitals (up to 200kPa – i.e high pressures)

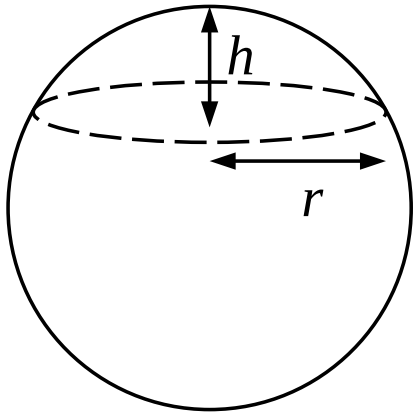




Diaphragm(膜气压计)

- **U**ses the deflection of a flexible membrane that separates regions of different pressure, with the deformation measured using mechanical, optical or capacitive techniques, range is to above 1.3 pa.

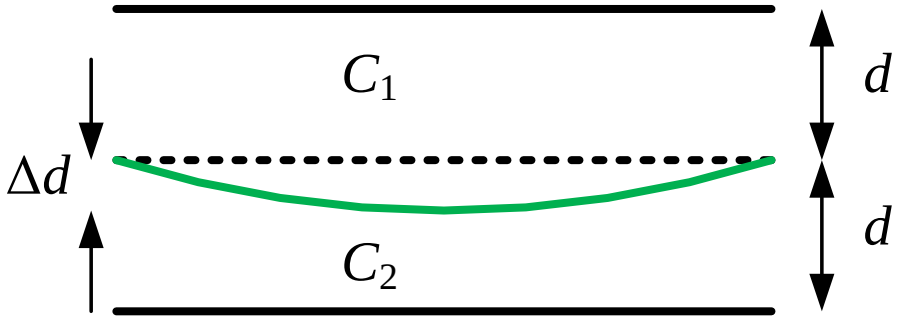


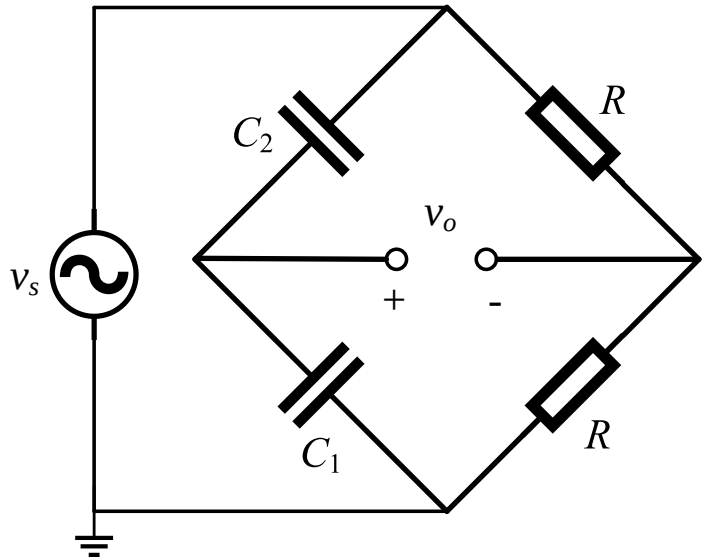


$$V = \frac{1}{6}\pi h\left(3r^2 + h^2\right)$$

Combine this with the capacitive methods

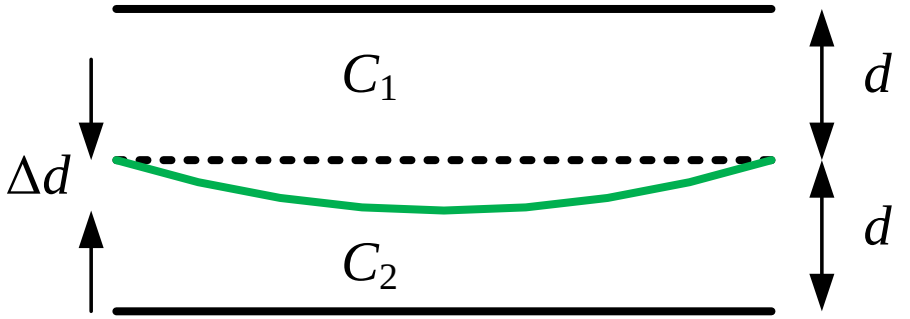
$$\frac{v_o}{v_s} = \frac{\Delta d}{2d}$$

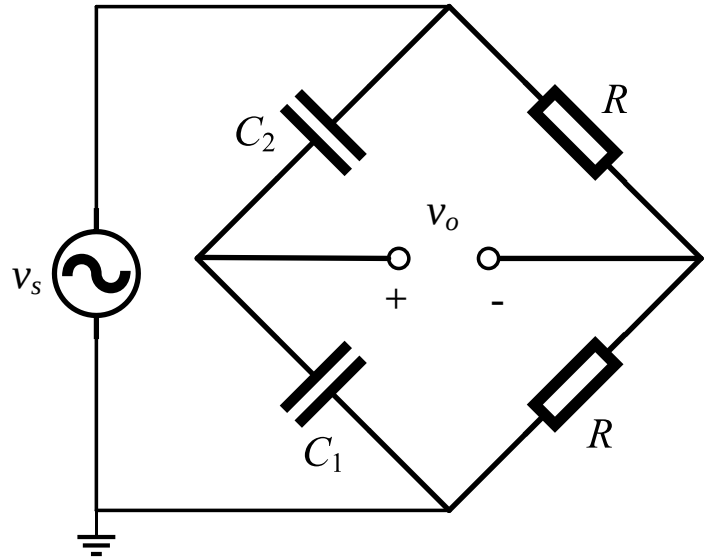




Class question

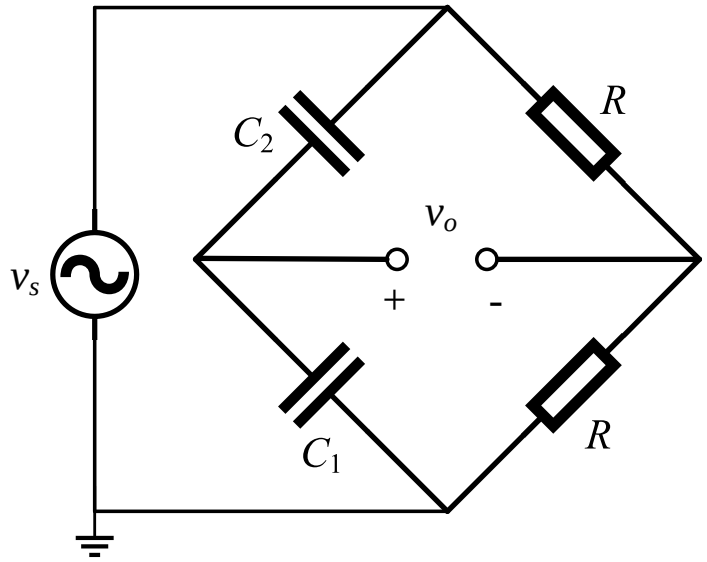
- Let's use example measurements with a chamber size of d= 1cm and the variable is Δd=0.01cm, the voltage supply to the bridge is Ac 1V and the output from the bridge is 0.237V





Answer

- The output of bridge is $\frac{v_o}{v_s} = \frac{\Delta d}{2d}$



$$\Delta d = \frac{v_o}{v_s} \times 2d$$

$$= \frac{0.237}{1} \times 2 \bullet 1 \times 10^{-2}$$

$$= 0.00474\text{cm}$$